

Supplementary Information

for “HPC-FLRNet: A Compact Deep Learning Model for Compressive Strength Prediction of High-Performance Concrete under Small Sample Conditions”

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Supplementary Note S1. HPC1826 construction and robust quality screening

This note provides the detailed, reproducible procedure used to construct the HPC1826 benchmark from three public sources, including harmonization, deduplication, and robust quality screening.

S1.1 Source datasets and harmonization

Three publicly available datasets were consolidated into a unified feature space. Feature names and physical units were aligned, and only the common mix-design descriptors and curing age were retained. After merging, the raw combined set contained 1,952 records. Exact duplicates were removed, yielding 1,925 unique records for subsequent screening.

During harmonization, source-specific fields not required by the unified 10-feature schema were excluded. In particular: data103—no additional fields were removed; data714—Tensile strength of cement fct (MPa), Splitting tensile strength fst,t (MPa), Slump (mm), Sand ratio (%), Fineness modulus of sand, Stone powder content in sand (%), and Dmax of crushed stone (mm); data1133—Superplasticizer (kg in a m³ mixture).

S1.2 Robust outlier screening (IQR-based; conservative intersection rule)

Because most mixture variables and strength exhibit skewness and heavy tails, we adopted a non-parametric interquartile range (IQR) screening approach rather than Z-score thresholds. For each numeric feature, records outside the standard IQR fences $[Q1 - 1.5 \cdot IQR, Q3 + 1.5 \cdot IQR]$ were flagged. To avoid excessive removal of rare-but-valid mixtures, a conservative “intersection” rule was used: a record was removed only when it was consistently flagged across multiple key variables, rather than by a single-feature trigger.

S1.3 Accounting of sample counts

After screening, 99 records (5.14%) were removed from the intermediate set ($1,925 \rightarrow 1,826$), producing the final HPC1826 benchmark.

Supplementary Note S2. Cross-source distribution overlap analysis

To evidence cross-source heterogeneity, we provide a source-aware PCA projection of the merged data. The visualization indicates partial overlap among sources alongside source-dependent shifts, motivating robust evaluation across multiple datasets.

Supplementary Note S3. Baseline configuration strategy and tuning ranges

Baselines were tuned using validation performance while keeping the test set untouched. The tuning strategy and candidate ranges are summarized in Table S3.

Supplementary Note S4. Additional interpretability visualizations (standardized dependence plots)

We provide additional SHAP dependence plots with each x-axis standardized (z-score). Standardization improves comparability across features with different physical units and reduces visual bias due to scale differences.

Supplementary Note S5. Statistical testing details and full post-hoc tables

Repeated-run experiments were conducted with 10 independent runs per model per dataset. The run index was treated as the blocking factor. We used the Friedman test to assess overall differences across methods, followed by paired Wilcoxon signed-rank post-hoc comparisons with Holm correction. Full CSV outputs are provided as supplementary data files.

Supplementary Tables

Table S1. HPC1826 construction accounting.

Stage	Records	Description
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Merged raw records	1952	After feature/unit alignment and merge of 3 public sources
After deduplication	1925	Exact duplicates removed
Final HPC1826	1826	After robust quality screening (IQR-based; conservative intersection rule)

Table S2. Input features in HPC1826 (aligned names and units).

Feature	Symbol (paper)	Unit
Cement content	Cem	kg/m ³
Slag content	Slg	kg/m ³
Fly ash content	FA	kg/m ³
Water content	Wat	kg/m ³
Water-to-binder ratio	W/B	ratio
Water-to-cement ratio	W/C	ratio
Superplasticizer-to-cement ratio	SP/C	ratio
Coarse aggregate content	CA	kg/m ³
Fine aggregate content	FAgg	kg/m ³
Curing age	Age	days

Table S3. Baseline tuning ranges (candidate grids used in scripts; include selected settings in main tables).

Model	Key tuned hyperparameters	Candidate values (grid)
SVR (RBF)	C, epsilon, gamma	$C \in \{5, 10, 20, 50, 100, 200, 500, 1000\};$ $\epsilon \in \{0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.07, 0.08, 0.09\};$ $\gamma \in \{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9\}$
GBRT	n_estimators, max_depth, learning_rate	$n_estimators \in \{100, 200, 300, 500, 1000\};$ $max_depth \in \{1, 2, 3, 4, 5, 6\};$ $lr \in \{0.005, 0.01, 0.02, 0.05, 0.1, 0.2\}$
XGBoost	n_estimators, max_depth, learning_rate	$n_estimators \in \{100, 200, 300, 500, 1000\};$ $max_depth \in \{1, 2, 3, 4, 5, 6\};$ $lr \in \{0.005, 0.01, 0.02, 0.05, 0.1, 0.2\}$

MLP	hidden_dim1, hidden_dim2, max_iter	hidden_dim1 $\in \{100, 200, 300, 400, 500\}$; hidden_dim2 $\in \{100, 200, 300, 400, 500\}$; max_iter $\in \{100, 200, 300, 500, 1000\}$
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Table S4. Friedman test summary and average ranks (RMSE; lower is better).

Dataset	Friedman χ^2	p-value	Best/Top ranks (avg rank)
data103	36.64	2.14×10^{-7}	HPC-FLRNet(1.0) < MLP(2.3) < SVR(2.7)
data714	26.96	2.03×10^{-5}	GBR(2.0) = XGB(2.0) $\approx$ HPC-FLRNet(2.2)
data1133	29.20	7.12×10^{-6}	HPC-FLRNet(1.7) $\approx$ XGB(2.0) < GBR(2.4)
HPC1826	33.04	1.17×10^{-6}	HPC-FLRNet(1.7) < GBR(1.9) < XGB(2.4)

Table S5. Ablation planned post-hoc (Full vs each variant; RMSE; Holm correction m=5).

Variant	Δ RMSE vs Full	% vs Full	p_raw	p_Holm
without CRM	0.8283	20.74%	0.00506	0.02531
without FRM	0.3444	8.62%	0.00506	0.02531
without IPM	0.2079	5.21%	0.00506	0.02531
without LayerNorm	0.1644	4.12%	0.00506	0.02531
without Dropout	0.2464	6.17%	0.00691	0.02531

Table S6. Ablation mean $\pm$ standard deviation (RMSE) on HPC1826.

Variant	RMSE (mean $\pm$ std)
Full	3.993 ± 0.055
without LayerNorm	4.158 ± 0.073
without IPM	4.201 ± 0.070
without Dropout	4.240 ± 0.140
without FRM	4.338 ± 0.103
without CRM	4.821 ± 0.049

Table S7. Split-wise performance of HPC-FLRNet on HPC1826.

Split	RMSE (mean $\pm$ std)	MAE (mean $\pm$ std)
Train	3.046 $\pm$ 0.226	2.192 $\pm$ 0.190
Validation	4.195 $\pm$ 0.127	3.041 $\pm$ 0.065
Test	4.388 $\pm$ 0.094	3.159 $\pm$ 0.101

Note: Split-wise performance of HPC-FLRNet on HPC1826 (training/validation/testing). Metrics are computed on the training subset (excluding the validation subset), the held-out validation subset, and the test subset under the fixed split protocol. R^2 (mean $\pm$ std): train 0.9725 ± 0.0041 ; validation 0.9471 ± 0.0032 ; test 0.9409 ± 0.0025 .

Fixed split protocol: HPC1826 is split into 82% train and 18% test, and 10% of the training portion is further held out as a validation set.

Supplementary Figures

Fig. S1. Source-aware PCA projection of HPC1826 (points colored by source) to evidence cross-source overlap and shifts.

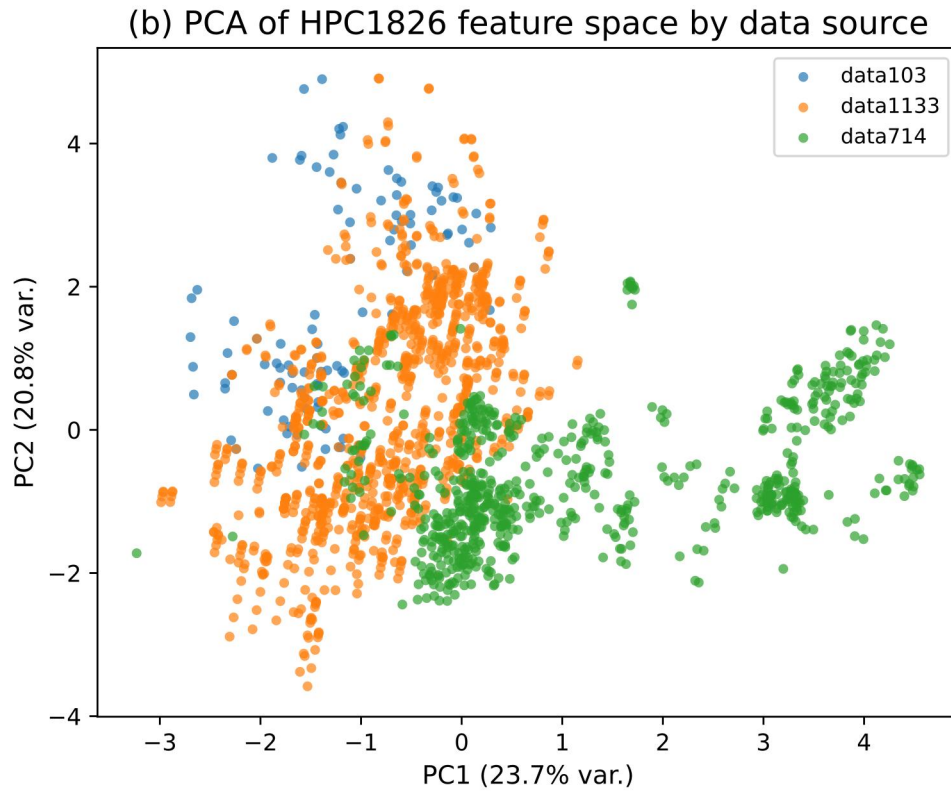


Fig. S2. Additional SHAP dependence plots (x-axis standardized as z-score) illustrating non-linear effects and interactions.

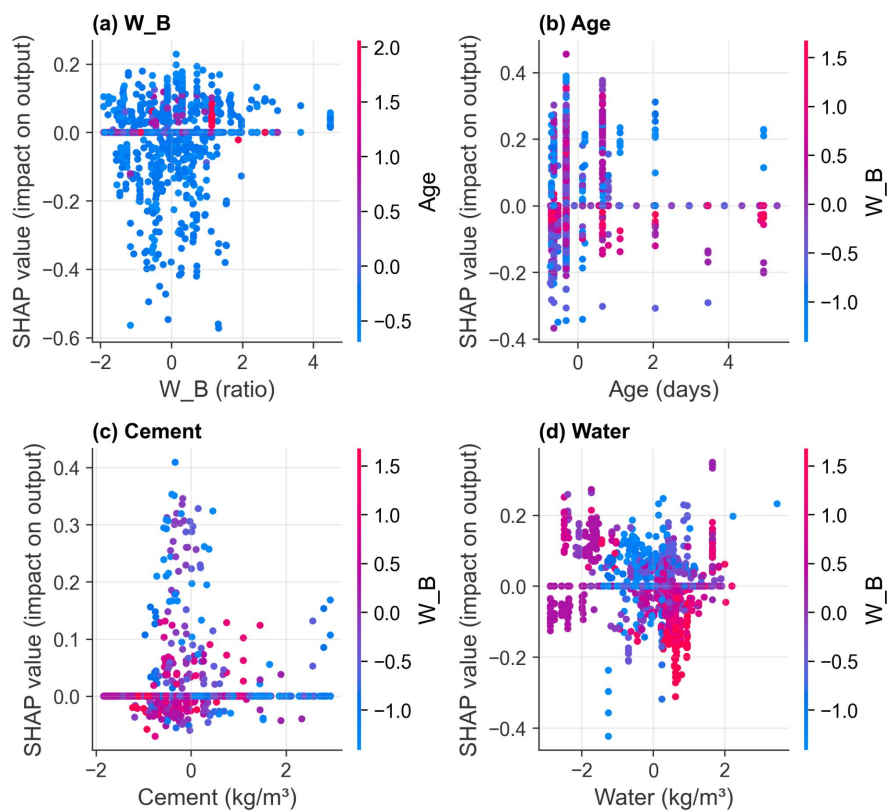


Fig. S3. SHAP global feature importance summary for HPC-FLRNet on HPC1826 (backup figure).

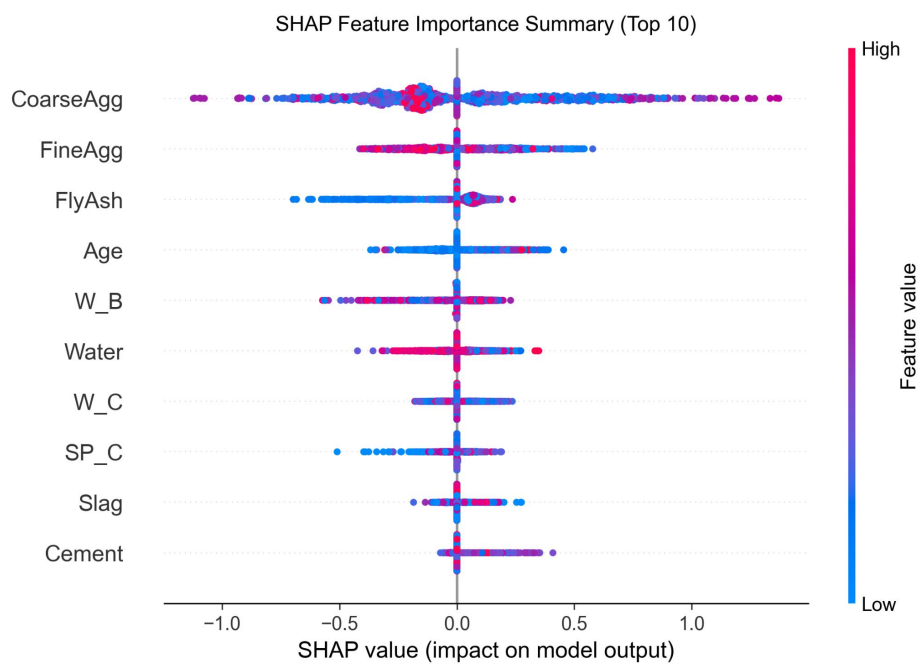


Fig. S4. Cross-check against classical water–binder empirical relations (Abrams/Bolomey-type trend) and model predictions.

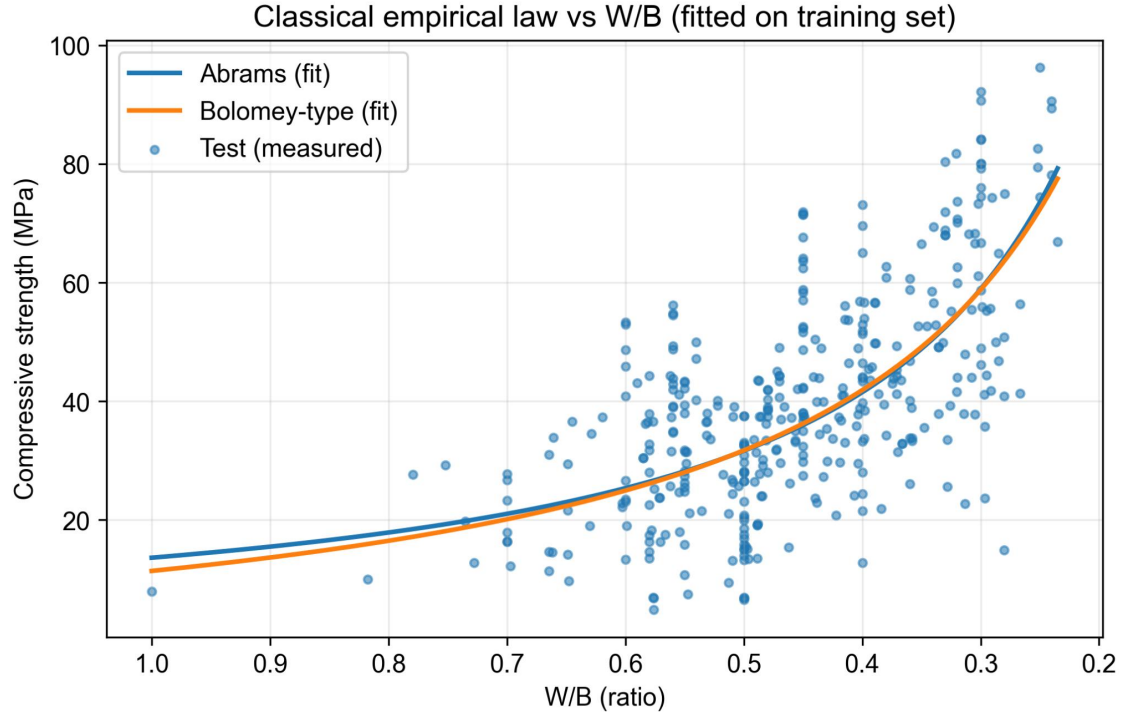
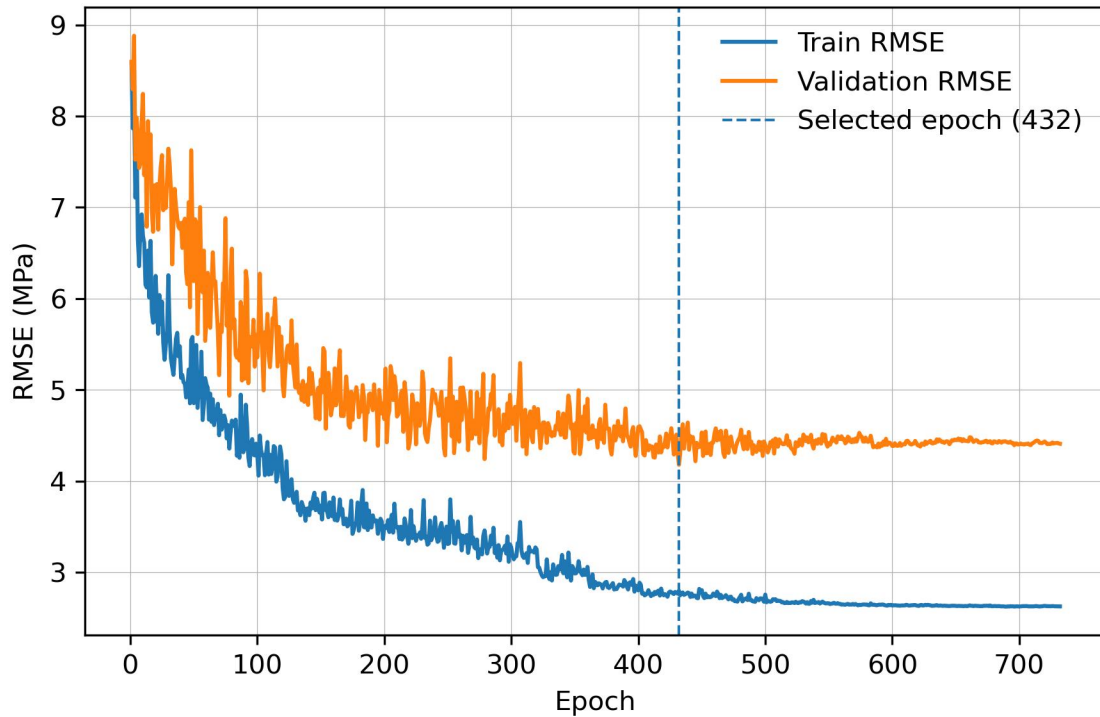


Fig. S5. Training and validation learning curves of HPC-FLRNet on HPC1826 under the fixed split. The selected checkpoint corresponds to the minimum validation loss (early_stop = 300).



Supplementary Data files (also available in the github repository)

- Supplementary Data 1 (data/intermediate/tmp_data1950_merged.xlsx): merged intermediate dataset after feature/unit alignment (before deduplication and screening).
- Supplementary Data 2 (data/intermediate/tmp_data1925_dedup.xlsx): intermediate dataset after deduplication.
- Supplementary Data 3 (data/processed/HPC1826.xlsx): final benchmark dataset (HPC1826) after robust quality screening.
- Supplementary Data 4–7 (results/significance/<dataset>/... and *results.json): statistical testing outputs for each dataset (Friedman summary, average ranks, Wilcoxon+Holm post-hoc tables; see per-folder CSV files).
- Supplementary Data 8 (results/ablation/hpc1826/ablation_posthoc_full_vs_each_holm.csv and *results.json): planned post-hoc tests (Full vs each ablation variant) with Holm correction and effect sizes.